WEB PRODUCTS · AI SYSTEMS · DEVELOPER PLATFORMS

Carlos Guerrero

Software Engineer | Web Products, AI & Platforms

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PROFILE

I build products and reusable systems that help teams deliver. My work spans React and TypeScript interfaces, real-time AI experiences, and deployment platforms that make infrastructure easier to use. I take ownership across product, backend, data, and infrastructure teams, adapting to what the company needs as products evolve.

EXPERIENCE

rPotential

March 2025 - August 2026

Software Engineer, AI Products & Platform

Joined early to build AI-powered decision-support products for executives, then expanded into shared infrastructure and integration across frontend, backend, and data teams.

- Built reusable frontend and backend templates adopted by three teams: frontend, backend, and data. Enabled teams to launch projects without managing the underlying deployment infrastructure.
- Built most initial prototypes in React and TypeScript, using Zustand for state management. Approximately half evolved into products before further iteration.
- Created real-time AI voice experiences that generated data presentations in response to user requests, addressing latency challenges to improve conversational responsiveness.
- Developed Slack-integrated AI for data gathering and user interaction, plus a Twilio and OpenAI Realtime telephone prototype to collect employee information and feedback.
- Connected frontend applications to evolving backends and data pipelines, collaborating on integrations requiring restricted source-data access and anonymized downstream information.

Anaconda

March 2024 - December 2025

Applied AI Engineer

Built serverless infrastructure for AI agents and LLM workloads, supporting deployment, orchestration, and integration through OpenAI-compatible interfaces.

- Created reusable deployment workflows and queue-based communication to support rapid agent iteration and asynchronous multi-agent workflows.
- Delivered a Slack-integrated agent connected to Jupyter, helping scientists improve notebooks and execute Python from Slack.
- Added tracing and monitoring with AWS X-Ray and CloudWatch to support performance analysis and debugging.

Masterworks

January 2018 - January 2020

Backend Engineer to DevOps Lead

Moved from backend engineering into DevOps leadership, building shared infrastructure to improve developer experience and deployment reliability.

- Created automated per-branch environments so teams could develop and validate changes independently.
- Migrated frontend hosting to AWS CloudFront, supporting CDN-level A/B testing and performance tuning.
- Reworked backend delivery from Jenkins and Kubernetes to GitHub Actions and AWS Fargate, with infrastructure managed through Terraform and AWS CDK.

SELECTED PLATFORMS & PROJECTS

W7S.io

Creator & Developer

w7s.io

Designed and built an open-source deployment platform that lets developers ship full-stack applications directly from GitHub, with infrastructure provisioned through a shared workflow.

- Unified frontend hosting, JavaScript/TypeScript backends, databases, storage, queues, and scheduled workflows so applications can share a consistent path from repository to deployment.
- Built GitHub Actions deployment with repository-based OIDC authentication, removing the need for separate deployment API keys.
- Integrated usage controls, operational visibility, and a read-only infrastructure API to help developers and AI agents inspect deployments and troubleshoot issues.

Democrachain

Creator & Developer | Independent platform and working prototype

github.com/democrachain

Designed and built a blockchain platform in JavaScript, later evolving it in TypeScript, to help organizations synchronize data across nodes and implement shared business logic through smart contracts.

- Combined synchronization, scoped permissions, and JavaScript smart contracts into a reusable foundation for distributed applications.
- Designed the platform to reduce shared-state coordination and cache-management work for application teams.

CacheP2P

Personal project

github.com/guerrerocarlos/CacheP2P

Built a browser-based peer-to-peer CDN using WebRTC, exploring how participating browsers can contribute capacity to content delivery.

TECHNOLOGIES & ENGINEERING

React · TypeScript · JavaScript · Zustand · Node.js · Python · REST APIs · Real-time AI · GitHub Actions · AWS · Cloudflare Workers · CI/CD · Distributed systems

EDUCATION

Universidad del Táchira

2002 - 2007

Electronic Engineering

Coursework: C++ · Process Automation · Digital Signal Processing · Control Systems · Microcontrollers · Electronics

ETI

2002 - 2019

Industrial Chemistry

Coursework: Chemical Engineering · Organic Chemistry · Inorganic Chemistry · Analytical Chemistry · Physical Chemistry · Biochemistry

TRAINING & CERTIFICATIONS

AWS Certified Cloud Practitioner

Amazon Web Services Training and Certification | 2020

Cisco Certified Network Associate (CCNA)

CISCO - Networking Academy | 2015

TEACHING & INSTRUCTION

- Delivered MongoDB courses as an instructor.
- Delivered Blender courses as an instructor.

LANGUAGES

English: fluent | Spanish: native